HANDWRITTEN NOTES

COMPLEX VARIABLES, PROBABILITY AND STATISTICS

Subject Code : 23A54403

UNIT - 1

Complex Variable - Differentiation

Topics covered in this unit :

- Introduction to functions of a complex variable
- Concept of limit and continuity
- Differentiation, Cauchy-Riemann equations
- Analytic functions, harmonic functions
- Finding harmonic conjugate
- Construction of analytic function: Milne-Thomson method

(Key terms are written in red , headings in purple/green, formulas on yellow.)

1. Complex Numbers - Quick Recap

$$z = x + iy \;,\; i^2 = -1 \;,\; x = Re(z) \;,\; y = Im(z)$$

Item	Formula
Conjugate	$\bar{z} = x - iy$
Modulus	$\lvert z\rvert = r = \sqrt{x^2 + y^2}$
Argument	$\theta = \arg z = \tan^{-1}(y/x)$ (principal value $-\pi < \theta \le \pi$)
Polar form	$z = r(\cos\theta + i\sin\theta) = r e^{i\theta}$
Product / quotient	$z_1 z_2 = r_1 r_2 e^{i(\theta_1+\theta_2)}$; $z_1/z_2 = (r_1/r_2) e^{i(\theta_1-\theta_2)}$
De Moivre	$(\cos t + i\sin t)^n = \cos nt + i\sin nt$
Euler	$e^{it} = \cos t + i\sin t$; $\cos t = (e^{it}+e^{-it})/2$; $\sin t = (e^{it}-e^{-it})/2i$
Triangle inequality	$\lvert z_1 + z_2\rvert \le \lvert z_1\rvert + \lvert z_2\rvert$

Useful points

- $z\,\bar{z} = \lvert z\rvert^2$; $1/z = \bar{z} / \lvert z\rvert^2$
- Argand plane: x-axis = real axis, y-axis = imaginary axis
- Neighbourhood of z_0: $\lvert z - z_0\rvert < \delta$ (open disc); region = open connected set
- $\lvert z - a\rvert = r$ is a circle ; $\lvert z - a\rvert < r$ is its interior

2. Functions of a Complex Variable

2.1 Definition

If to each complex number $z = x + iy$ of a set S there corresponds one (or more) complex number $w = u + iv$, then w is a function of z : $w = f(z) = u(x, y) + i\,v(x, y)$, where u and v are real functions of x and y.

- Single valued: one w for each z (e.g. z^2, e^z). Multi-valued: more than

one w (e.g. z^(1/2), log z).

Examples

f(z)	u(x,y)	v(x,y)
z^2	x^2 - y^2	2xy
z^3	x^3 - 3xy^2	3x^2 y - y^3
e^z	e^x cos y	e^x sin y
sin z	sin x cosh y	cos x sinh y
cos z	cos x cosh y	- sin x sinh y
1/z	x/(x^2+y^2)	-y/(x^2+y^2)
z-bar	x	-y
|z|^2	x^2 + y^2	0

2.2 Elementary functions

Function	Definition / property
Exponential	e^z = e^x (cos y + i sin y); periodic with period 2 pi i; e^z != 0
Trigonometric	sin z = (e^(iz) - e^(-iz))/2i ; cos z = (e^(iz) + e^(-iz))/2
Hyperbolic	sinh z = (e^z - e^-z)/2 ; cosh z = (e^z + e^-z)/2 ; sin(iz) = i sinh z
Logarithm	log z = log r + i (theta + 2 n pi); principal value Log z = log r + i theta
Power	z^a = e^(a log z)

Remember

- sin^2 z + cos^2 z = 1 holds for complex z; but \|sin z\| and \|cos z\| can exceed 1
- cosh^2 z - sinh^2 z = 1; cos(iy) = cosh y; sin(iy) = i sinh y
- log(-1) = i pi (principal); general: (2n + 1) pi i

3. Limit and Continuity

3.1 Limit

lim f(z) = l as z -> z0 means: for every epsilon > 0 there is a delta > 0 such that |f(z) - l| < epsilon whenever 0 < |z - z0| < delta .

Key difference from real limit

- In real calculus we approach x0 only from left and right (2 ways)
- In complex plane z can approach z0 along infinitely many paths (line, curve, spiral)
- Limit exists only if it is the SAME along all paths ; if two paths give different values -> limit does not exist

Properties

If lim f = l and lim g = m, then lim (f +/- g) = l +/- m ; lim (f g) = l m ; lim (f/g) = l/m (m != 0).

Example 1 - Limit by definition

Show that lim z^2 = 4 as z -> 2. Take z = 2 + h: |z^2 - 4| = |h||h+4| < 5|h| for |h| < 1. Choose delta = min(1, epsilon/5). Done.

Example 2 - Limit does not exist

f(z) = x y / (x^2 + y^2) at z -> 0. Along y = m x: f = m x^2 / (x^2 (1 + m^2)) = m/(1+m^2) - depends on m (slope). Hence limit does not exist .

Example 3

f(z) = z-bar / z at z -> 0. Along real axis (y=0): f = x/x = 1. Along imaginary axis (x=0): f = (-iy)/(iy) = -1. Different -> limit does not exist .

3.2 Continuity

f(z) is continuous at z0 <=> lim f(z) (z -> z0) = f(z0)

i.e. (1) f(z0) defined, (2) limit exists, (3) both are equal. f = u + iv is continuous iff u(x,y) and v(x,y) are both continuous. Sums, products, quotients (denominator != 0) and composites of continuous functions are continuous.

Polynomials, e^z, sin z, cos z are continuous everywhere.

Example: f(z) = (z^2 + 1)/(z - i)... define f(i) = 2i to make it continuous (removable) since z^2+1 = (z-i)(z+i).

4. Differentiation - Derivative of f(z)

$$f'(z_0) = \lim_{h \to 0} \left[f(z_0 + h) - f(z_0) \right] / h$$

The limit must be independent of the way h -> 0 (h complex). If it exists, f is differentiable at z_0. All usual rules (sum, product, quotient, chain) and formulas hold:

f(z)	f'(z)
z^n	n z^(n-1)
e^z	e^z
sin z, cos z	cos z, - sin z
sinh z, cosh z	cosh z, sinh z
log z	1/z
tan z	sec^2 z

Important facts

- Differentiable at a point => continuous there (converse false)
- f(z) = z-bar, |z|^2, Re(z), Im(z) are continuous everywhere but nowhere (or almost nowhere) differentiable
- Complex differentiability is much stronger than real differentiability

5. Cauchy-Riemann Equations

5.1 Derivation

Let $f(z) = u + iv$ be differentiable at $z = x + iy$. Take $h \to 0$ along the real axis ($h = dx$):

$$f'(z) = u_x + i\, v_x$$

Take $h \to 0$ along imaginary axis ($h = i\, dy$):

$$f'(z) = (1/i)(u_y + i\, v_y) = v_y - i\, u_y$$

Both limits must be equal. Comparing real and imaginary parts gives the C-R equations :

Cauchy-Riemann (Cartesian form)

$$u_x = v_y \text{ and } u_y = -v_x$$

Meaning: $u_x = du/dx$ etc. The derivative can then be written $f'(z) = u_x + i\, v_x = v_y - i\, u_y$.

5.2 Polar form of C-R equations

For $z = r\, e^{(i\, theta)}$, $f = u(r, theta) + i\, v(r, theta)$:

$$u_r = (1/r)\, v_{theta} \text{ and } v_r = -(1/r)\, u_{theta}$$

$$f'(z) = e^{(-i\, theta)}\, (\, u_r + i\, v_r\,)$$

5.3 Sufficient condition

If u, v, u_x, u_y, v_x, v_y are continuous in a region and satisfy the C-R equations, then $f(z)$ is differentiable (analytic) there. (C-R equations alone are necessary but not sufficient .)

Example 1 - $f(z) = z^2$

$u = x^2 - y^2$, $v = 2xy$. $u_x = 2x$, $u_y = -2y$, $v_x = 2y$, $v_y = 2x$. So $u_x = v_y$ and $u_y = -v_x$ -> C-R satisfied everywhere, $f'(z) = 2x + i\, 2y = 2z$.

Example 2 - f(z) = z-bar

u = x, v = -y: u_x = 1, v_y = -1 -> u_x != v_y. C-R not satisfied -> not differentiable anywhere.

Example 3 - f(z) = e^z

u = e^x cos y, v = e^x sin y: u_x = e^x cos y = v_y ; u_y = - e^x sin y = - v_x. OK. f'(z) = u_x + i v_x = e^x (cos y + i sin y) = e^z .

Example 4 - f(z) = |z|^2

u = x^2 + y^2, v = 0: u_x = 2x, v_y = 0; u_y = 2y, v_x = 0. C-R holds only at origin (x = y = 0). So differentiable only at z = 0, nowhere analytic .

Example 5 - Show f(z) = sinh z is analytic

sinh z = sinh x cos y + i cosh x sin y. u_x = cosh x cos y = v_y ; u_y = - sinh x sin y = - v_x. C-R satisfied everywhere -> entire .

6. Analytic Functions

6.1 Definition

A function f(z) is analytic (holomorphic, regular) at a point z0 if it is differentiable at z0 and at every point in some neighbourhood of z0 . f is analytic in a region R if it is analytic at every point of R. An entire function is analytic in the whole plane (polynomials, e^z, sin z, cos z).

Differentiable at a point	Analytic at a point
Only at that point is enough	Needs differentiability in a whole neighbourhood
|z|^2 is differentiable at 0	|z|^2 is analytic nowhere
Local property	Stronger property

6.2 Singular points

A point where f(z) fails to be analytic is a singular point . Example: f(z) = 1/(z - 2): singular point z = 2; analytic elsewhere. f(z) = (z + 1)/((z-1)(z+3)): singular at z = 1 and z = -3.

Properties of analytic functions

- Sum, difference, product and composite of analytic functions are analytic, quotient analytic where denominator != 0
- If f is analytic then f'(z), f''(z), ... all exist and are analytic (infinitely differentiable)
- If f = u + iv is analytic, u and v are harmonic
- Curves u = c1 and v = c2 are orthogonal (cut at right angles)
- If f is analytic and f' = 0 everywhere in a region, f is constant
- If f and f-bar are both analytic, f is constant, if |f| is constant, f is constant

Orthogonal trajectories

Slope of u = c1 is $-u_x/u_y$, slope of v = c2 is $-v_x/v_y$. Product = $(u_x\ v_x)/(u_y\ v_y) = (u_x\ v_x)/((-v_x)(u_x)) = -1$ (using C-R). Hence orthogonal.

7. Harmonic Functions

7.1 Laplace equation

A real function phi(x, y) with continuous second partial derivatives is harmonic if it satisfies Laplace equation :

$$\phi_{xx} + \phi_{yy} = 0 \quad (\text{i.e. } \nabla^2 \phi = 0)$$

7.2 Theorem

If f(z) = u + iv is analytic, then u and v are both harmonic. Proof: C-R: $u_x = v_y$, $u_y = -v_x$. Differentiate: $u_{xx} = v_{yx}$, $u_{yy} = -v_{xy}$. Add: $u_{xx} + u_{yy} = v_{yx} - v_{xy} = 0$ (mixed partials are equal). Similarly $v_{xx} + v_{yy} = 0$.

Harmonic conjugate

- If u and v are harmonic and satisfy C-R ($u_x = v_y$, $u_y = -v_x$), then v is the harmonic conjugate of u
- If v is conjugate of u, then u is conjugate of -v (not of v!)
- Not every pair of harmonic functions is a conjugate pair: e.g. $u = x^2 - y^2$ and $v = x^2 - y^2$ are harmonic but not conjugates

Example 1 - Show $u = x^2 - y^2$ is harmonic

$u_x = 2x$, $u_{xx} = 2$; $u_y = -2y$, $u_{yy} = -2$. $u_{xx} + u_{yy} = 0$ -> harmonic .

Example 2 - Show $u = e^x \cos y$ is harmonic

$u_{xx} = e^x \cos y$; $u_{yy} = -e^x \cos y$; sum = 0. Harmonic.

Example 3 - Is $u = x^3 - 3xy^2 + 3x^2 - 3y^2$ harmonic?

$u_{xx} = 6x + 6$, $u_{yy} = -6x - 6$. Sum = 0 -> harmonic .

Example 4 - Is $u = x^2 + y^2$ harmonic?

$u_{xx} + u_{yy} = 2 + 2 = 4 \neq 0$ -> not harmonic . Hence no analytic function has this as real part.

8. Finding the Harmonic Conjugate

8.1 Direct (partial integration) method

1. Verify the given u is harmonic
2. Use C-R: $dv = v_x\,dx + v_y\,dy = -u_y\,dx + u_x\,dy$
3. Integrate (the form is exact since u is harmonic) -> $v = \ldots + C$

$$v = \int (-u_y\,dx + u_x\,dy) \text{ (exact differential)}$$

Example 1

Find the harmonic conjugate of $u = x^2 - y^2 + x$. Here $u_x = 2x + 1$, $u_y = -2y$.

$dv = -u_y\,dx + u_x\,dy = 2y\,dx + (2x + 1)\,dy$. Integrate: $v = 2xy + y + C$

[check: $v_x = 2y$, $v_y = 2x + 1$ OK].

So $f(z) = (x^2 - y^2 + x) + i(2xy + y + C) = z^2 + z + iC$.

Example 2

$u = e^x \cos y$. $u_x = e^x \cos y$, $u_y = -e^x \sin y$. $dv = e^x \sin y\, dx + e^x \cos y\, dy = d(e^x \sin y)$. $v = e^x \sin y + C$, and $f = e^z + iC$.

Example 3 - Given v find u

$v = 3x^2 y - y^3$. $v_x = 6xy$, $v_y = 3x^2 - 3y^2$. $du = u_x\, dx + u_y\, dy = v_y\, dx - v_x\, dy = (3x^2 - 3y^2)\, dx - 6xy\, dy$. Integrating: $u = x^3 - 3xy^2 + C$. $f = z^3 + C$.

9. Construction of Analytic Function -

Given u (or v), find $f(z)$ directly in terms of z without finding the conjugate first.

Milne-Thomson Method

- If u is given: find u_x and u_y. Then $f'(z) = u_x - i\, u_y$ and put $x = z$, $y = 0$
- $f'(z) = u_x(z, 0) - i\, u_y(z, 0)$ -> integrate w.r.t. z: $f(z) = \int [u_x(z,0) - i\, u_y(z,0)]\, dz + C$
- If v is given: $f'(z) = v_y + i\, v_x$ and put $x = z$, $y = 0$
- $f(z) = \int [\, v_y(z,0) + i\, v_x(z,0)\,]\, dz + C$
- C is a complex constant (found from a given condition such as $f(0)$ or $f(1+i)$)

Why does it work?

$x = (z + \bar{z})/2$, $y = (z - \bar{z})/2i$. Since $f(z) = u + iv$ is a function of z only, and identity holds for all $\bar{z}$, put $\bar{z} = z$, i.e. $x = z$, $y = 0$.

Example 1 - $u = x^3 - 3xy^2 + 3x^2 - 3y^2 + 1$

$u_x = 3x^2 - 3y^2 + 6x$; $u_y = -6xy - 6y$. Put $x = z$, $y = 0$: $u_x = 3z^2 + 6z$, $u_y = 0$.

$f'(z) = u_x - i\, u_y = 3z^2 + 6z$. Integrate: $f(z) = z^3 + 3z^2 + C$. (Constant

u has + 1 so C = 1 + ic.)

Example 2 - u = e^x (x cos y - y sin y)

u_x = e^x (x cos y - y sin y) + e^x cos y ; u_y = e^x (- x sin y - sin y - y cos y).

At (z, 0): u_x = e^z (z + 1) ; u_y = 0. f'(z) = (z + 1) e^z. Integrate by parts: f(z) = z e^z + C.

Example 3 - v = e^x (x sin y + y cos y); find f(z)

v_x = e^x (x sin y + y cos y) + e^x sin y ; v_y = e^x (x cos y + cos y - y sin y). At (z,0): v_x = 0, v_y = e^z (z + 1).

f'(z) = v_y + i v_x = e^z (z + 1) -> f(z) = z e^z + C .

Example 4 - u - v given (important type)

If u - v = (x - y)(x^2 + 4xy + y^2), find f(z) = u + iv. Let F = (1+i) f = (u - v) + i (u + v). Then U = u - v (real part of analytic F). U_x = 3x^2 + 6xy - 3y^2... computing: U = x^3 + 3x^2 y - 3x y^2 - y^3 ; U_x = 3x^2 + 6xy - 3y^2 ; U_y = 3x^2 - 6xy - 3y^2. At (z,0): U_x = 3z^2, U_y = 3z^2. F' = U_x - i U_y = 3(1 - i) z^2 -> F = (1 - i) z^3 + C. So f = F/(1+i) = ((1-i)/(1+i)) z^3 = -i z^3 + C' .

Given	Use	Formula
u	Milne-Thomson	f' = u_x - i u_y at (z, 0)
v	Milne-Thomson	f' = v_y + i v_x at (z, 0)
u + v or u - v	Consider (1 + i) f or (1 - i) f	(1+i) f = (u - v) + i (u + v)
Polar u(r,theta)	Use polar C-R	f' = e^(-i theta)(u_r + i v_r), put r = z, theta = 0

Quick revision - Unit I

- $w = f(z) = u + iv$, limit must be same along all paths
- C-R: $u_x = v_y$, $u_y = -v_x$; polar: $u_r = v_\theta / r$, $v_r = -u_\theta / r$
- Analytic = differentiable in a neighbourhood, u, v harmonic: $u_{xx} + u_{yy} = 0$
- Harmonic conjugate: $v = \int(-u_y\,dx + u_x\,dy)$
- Milne-Thomson: $f' = u_x - i\,u_y$ $(x = z, y = 0)$ or $v_y + i\,v_x$

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COMPLEX VARIABLES, PROBABILITY AND STATISTICS

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UNIT - II

Complex Variable - Integration

Topics covered in this unit :

- Line integral, contour integration
- Cauchy's integral theorem (simple case)
- Cauchy's integral formula
- Power series: Taylor's series, zeros of analytic functions
- Singularities, Laurent's series
- Residues, Cauchy residue theorem (without proof)
- Evaluation of definite integrals involving sine and cosine

(Key terms are written in red , headings in purple/green, formulas on yellow.)

1. Line (Contour) Integral

1.1 Definition

Let C be a smooth curve from A to B given by $z = z(t) = x(t) + i\,y(t)$, $a <= t <= b$, and $f(z) = u + iv$ continuous on C. The line integral of f along C is:

$$\int_C f(z)\,dz = \int (u + iv)(dx + i\,dy) = \int (u\,dx - v\,dy) + i \int (v\,dx + u\,dy)$$

Parametric form

$$\int_C f(z)\,dz = \int_{t=a}^{b} f(z(t))\,z'(t)\,dt$$

- A contour is a piecewise smooth curve, a simple closed contour does not cross itself. Positive direction = anticlockwise.
- Integral over a closed contour is written as contour-integral (circle on the integral sign).

1.2 Properties

Property	Statement
Linearity	$\int (a f + b g)\,dz = a \int f\,dz + b \int g\,dz$
Reversal	$\int_{-C} f\,dz = -\int_C f\,dz$
Additivity	$\int_{C1+C2} = \int_{C1} + \int_{C2}$
ML inequality	$\left|\int_C f\,dz\right| <= M L$, where $\|f\| <= M$ on C and L = length of C

1.3 Useful parametrisations

Path	Parametrisation
Circle $\|z - a\| = r$	$z = a + r e^{i t}$, $0 <= t <= 2\pi$, $dz = i r e^{it}\,dt$
Line from z_1 to z_2	$z = z_1 + t(z_2 - z_1)$, $0 <= t <= 1$
Real axis segment	$z = x$, $dz = dx$
Parabola $y = x^2$	$z = x + i x^2$, $dz = (1 + 2 i x)\,dx$

Example 1

Evaluate $\int z^2\,dz$ from 0 to $1+i$ along the line $y = x$. Put $z = (1+i)t$, $dz = (1+i)\,dt$, $0 \le t \le 1$:

$\int = \int (1+i)^3 t^2\,dt = (1+i)^3/3 = (1+i)(2i)/3 = (2i-2)/3 = (-2+2i)/3$.

Example 2

Evaluate $\int \bar{z}\,dz$ from 0 to $1+i$ along $y = x$. $\bar{z} = (1-i)t$, $dz = (1+i)\,dt$: $\int = (1-i)(1+i)\int t\,dt = 2 \times 1/2 = 1$. Along a different path (x-axis then vertical) the value is $1+i$: depends on the path because $\bar{z}$ is not analytic.

Example 3 - the basic circle integral

> contour $\int$ over $|z-a| = r$ of $dz/(z-a)^n = 2\pi i$ $(n = 1)$; 0 $(n \neq 1$, integer$)$

Put $z - a = r\,e^{it}$: $dz = i\,r\,e^{it}\,dt$, so $\int = i\,r^{1-n}\int_0^{2\pi} e^{i(1-n)t}\,dt$. For $n = 1$ this is $i \times 2\pi = 2\pi i$; for $n \neq 1$ the integral of e^{ikt} over a full period is 0.

2. Cauchy's Integral Theorem

2.1 Statement (simple case)

Cauchy's Theorem (Cauchy-Goursat)

- If $f(z)$ is analytic at all points inside and on a simple closed contour C, then
- contour $\int_C f(z)\,dz = 0$
- Proof idea (Green's theorem): $\int (u\,dx - v\,dy) = \iint (-v_x - u_y)\,dA = 0$ and $\int (v\,dx + u\,dy) = \iint (u_x - v_y)\,dA = 0$ by C-R equations

2.2 Consequences

- Independence of path: if f is analytic in a simply connected region, Int from a to b of f dz depends only on a and b, not on the path
- Existence of antiderivative: Int from a to b f(z) dz = F(b) - F(a) where F' = f
- Deformation of contours: C1 can be deformed to C2 without changing the integral provided no singularity is crossed
- Multiply connected region (with holes): integral over outer C = sum of integrals over inner circles C1, C2, ...

Examples

(i) Int_C e^z dz = 0 for any closed C (e^z entire). (ii) Int_C z^3 dz = 0. (iii) Int_C dz/(z - 3) over |z| = 1 = 0 because singular point z = 3 lies outside C. (iv) Int_C 1/(z^2 + 4) dz over |z| = 1: singular points z = +/- 2i are outside |z| = 1 -> 0 .

3. Cauchy's Integral Formula

Cauchy's Integral Formula

- If f(z) is analytic inside and on a simple closed contour C and a is any point inside C, then
- f(a) = (1 / 2 pi i) contour Int_C f(z) / (z - a) dz
- i.e. contour Int_C f(z)/(z - a) dz = 2 pi i f(a)

Derivative form (Generalised formula)

f^(n)(a) = (n! / 2 pi i) contour Int_C f(z) / (z - a)^(n+1) dz

contour Int_C f(z)/(z - a)^(n+1) dz = 2 pi i f^(n)(a) / n!

The values of an analytic function inside a contour are determined by its values on the contour. Consequence: analytic function has derivatives of all

orders .

Example 1

Evaluate contour Int_C e^(2z)/(z - 1) dz, C: |z| = 2. Here a = 1 (inside C), f(z) = e^(2z). Integral = 2 pi i f(1) = 2 pi i e^2 .

Example 2 - using partial fractions

Evaluate contour Int_C dz/((z - 1)(z - 3)), C: |z| = 2. Partial fractions: 1/((z-1)(z-3)) = (1/2)[1/(z-3) - 1/(z-1)]. Only z = 1 is inside |z| = 2, so Int = (1/2)(0 - 2 pi i) = - pi i .

Example 3 - derivative form

Evaluate contour Int_C e^z/(z - 1)^3 dz, C: |z| = 3. Here n = 2, f(z) = e^z, f''(1) = e. Integral = 2 pi i f''(1)/2! = pi i e .

Example 4

Evaluate contour Int_C z/(z^2 + 1) dz, C: |z| = 2. Partial fractions: z/((z - i)(z + i)) = (1/2)[1/(z - i) + 1/(z + i)]. Both poles lie inside, so Int = (1/2)(2 pi i + 2 pi i) = 2 pi i .

4. Power Series - Taylor's Series

4.1 Sequences and series

A series Sum a_n (z - a)^n converges for |z - a| < R (radius of convergence R) and diverges for |z - a| > R. Ratio test: R = lim |a_n / a_(n+1)|.

4.2 Taylor's theorem

Taylor's Series

- If f(z) is analytic inside the circle C: |z - a| = R, then for all z inside C
- f(z) = f(a) + f'(a)(z - a) + f''(a)(z - a)^2 / 2! + ... = Sum (n = 0 to inf) f^(n)(a) (z - a)^n / n!
- When a = 0 it is called Maclaurin's series
- Radius of convergence = distance from a to the nearest singularity

4.3 Standard expansions (Maclaurin)

Function	Series	Valid for
e^z	1 + z + z^2/2! + z^3/3! + ...	all z
sin z	z - z^3/3! + z^5/5! - ...	all z
cos z	1 - z^2/2! + z^4/4! - ...	all z
sinh z	z + z^3/3! + z^5/5! + ...	all z
cosh z	1 + z^2/2! + z^4/4! + ...	all z
1/(1 - z)	1 + z + z^2 + z^3 + ...	|z| < 1
1/(1 + z)	1 - z + z^2 - z^3 + ...	|z| < 1
log(1 + z)	z - z^2/2 + z^3/3 - ...	|z| < 1
(1 + z)^n	1 + n z + n(n-1)/2! z^2 + ...	|z| < 1

Example 1 - expand f(z) = 1/(z + 2) about z = 1

Write z + 2 = 3 + (z - 1) = 3 [1 + (z - 1)/3]. Then f = (1/3) Sum (-1)^n ((z - 1)/3)^n valid for |z - 1| < 3 (nearest singularity z = -2 at distance 3).

Example 2 - expand sin z about z = pi/4

f = sin z, f' = cos z ...; f(pi/4) = f'(pi/4) = 1/sqrt2, f'' = - f ...; so sin z = (1/sqrt2)[1 + (z - pi/4) - (z - pi/4)^2/2! - (z - pi/4)^3/3! + ...].

4.4 Zeros of an analytic function

A point z = a is a zero of f(z) if f(a) = 0. It is a zero of order (multiplicity) m if f(a) = f'(a) = ... = f^(m-1)(a) = 0 but f^(m)(a) != 0, equivalently f(z) = (z - a)^m g(z), with g analytic and g(a) != 0.

Zero of order 1 is a simple zero . Zeros of a non-constant analytic function are isolated .

Function	Zero	Order
$z^2 - 1$	$z = +/- 1$	1 (simple)
$z^3 (z - 2)$	$z = 0$; $z = 2$	3 ; 1
sin z	$z = n\ pi$	1
$1 - \cos z$	$z = 0$	2 ($1 - \cos z = z^2/2 - \ldots$)
$z - \sin z$	$z = 0$	3 ($= z^3/6 - \ldots$)

5. Singularities

A point $z = a$ where $f(z)$ is not analytic is a singular point . If f is analytic in some neighbourhood of a except at a itself, a is an isolated singularity .

Type	Definition (from Laurent series)	Example
Removable	No negative powers; lim f(z) exists finite	sin z / z at z = 0 (value 1)
Pole of order m	Finite number (m) of negative powers; lim |f| = inf	$1/(z-1)^2$ at z = 1 (order 2)
Simple pole	Pole of order 1	$1/(z - 2)$ at z = 2
Essential	Infinite number of negative powers, limit does not exist	$e^{(1/z)}$, $\sin(1/z)$ at z = 0

How to find the order of a pole

- f(z) = phi(z)/(z - a)^m with phi(a) != 0 -> pole of order m
- Or: if f = g/h and h has a zero of order m at a while g(a) != 0 -> pole of order m
- Branch point (log z, sqrt z) and non-isolated singularities are other kinds
- Entire function: no finite singularity; singularity at infinity decided by f(1/w) at w = 0

Examples

(i) f = z/(z - 1)^2 (z + 2): poles z = 1 (order 2), z = -2 (simple). (ii) f = (1 - cos z)/z^2: z = 0 removable (limit 1/2). (iii) f = 1/sin z: simple poles at n pi. (iv) f = z e^(1/z): essential singularity at 0.

6. Laurent's Series

Laurent's theorem

- If f(z) is analytic in the annulus r1 < |z - a| < r2, then for all z in it
- f(z) = Sum (n = 0 to inf) a_n (z - a)^n + Sum (n = 1 to inf) b_n / (z - a)^n
- a_n = (1/2 pi i) contour Int f(z)/(z - a)^(n+1) dz ; b_n = (1/2 pi i) contour Int f(z)(z - a)^(n-1) dz
- First part = analytic (regular) part ; second part = principal part

- Principal part has no terms -> removable singularity; finite terms -> pole; infinite terms -> essential singularity.
- In practice we do not use the integrals: expand with standard series + binomial / geometric in the correct region.

Method

1. Mark singular points and draw the region (annulus) asked
2. Partial fractions if rational
3. For |z| < a use 1/(1 - z/a) = Sum (z/a)^n ; for |z| > a use 1/(z - a) = (1/z) Sum (a/z)^n
4. Collect powers of (z - a)

Example 1

Expand f(z) = 1/(z^2 - z) in (i) 0 < |z| < 1, (ii) |z| > 1. Partial fractions: 1/(z(z-1)) = 1/(z - 1) - 1/z.

(i) 0 < |z| < 1: 1/(z - 1) = - (1 + z + z^2 + ...), so f = - 1/z - 1 - z - z^2 - ...

(ii) |z| > 1: 1/(z - 1) = (1/z)(1 + 1/z + 1/z^2 + ...), so f = 1/z^2 + 1/z^3 + 1/z^4 + ...

Example 2 - f(z) = e^z / z^2 about z = 0

e^z = 1 + z + z^2/2! + ... so f = 1/z^2 + 1/z + 1/2 + z/6 + ... Pole of order 2; coefficient of 1/z = 1 (residue).

Example 3 - f(z) = 1/((z-1)(z-2)) in 1 < |z| < 2

f = 1/(z - 2) - 1/(z - 1). For |z| < 2: 1/(z - 2) = -(1/2) Sum (z/2)^n. For |z| > 1: 1/(z - 1) = Sum 1/z^(n+1).

f = - Sum (n=0 to inf) z^n / 2^(n+1) - Sum (n=0 to inf) 1/z^(n+1) .

Example 4 - sin z / z^3

= (z - z^3/6 + ...)/z^3 = 1/z^2 - 1/6 + z^2/120 - ... ; pole of order 2.

7. Residues and the Residue Theorem

7.1 Definition

If a is an isolated singularity of f with Laurent series about a, the residue of f at a is the coefficient b_1 of 1/(z - a) :

Res f(z) at z = a = b_1 = (1/2 pi i) contour Int_C f(z) dz

7.2 Formulae for residues

Case	Residue at z = a
Simple pole	lim (z -> a) (z - a) f(z)
f = P(z)/Q(z), Q(a) = 0, Q'(a) != 0	P(a) / Q'(a)
Pole of order m	(1/(m-1)!) lim (z -> a) d^(m-1)/dz^(m-1) [(z - a)^m f(z)]
Pole of order 2	lim d/dz [(z - a)^2 f(z)]
Removable singularity	0
Essential	Find coefficient of 1/(z - a) in Laurent series

Examples

(i) $f = 1/(z^2 + 1)$ at $z = i$: simple pole. Res = lim $(z - i)/((z - i)(z + i)) = 1/(2i)$.

(ii) $f = e^z/(z - 1)^2$ at $z = 1$: order 2. Res = $d/dz\ (e^z)$ at $1 = e$.

(iii) $f = z/(z^2 + 1)$... P/Q' rule: at $z = i$: $i/(2i) = 1/2$. (iv) $f = \cot z$ at $z = 0$: $\cos z / (\sin z)' = \cos 0/\cos 0 = 1$.

(v) $f = 1/(z^3 (z - 1))$ at $z = 0$ order 3: $(1/2!)\ d^2/dz^2\ [1/(z - 1)]$ at $0 = (1/2)(2/(z-1)^3) = -1$.

7.3 Cauchy's Residue Theorem (without proof)

Residue Theorem

- If f(z) is analytic inside and on a simple closed contour C except for a finite number of isolated singularities a1, a2, ..., an inside C, then
- contour Int_C f(z) dz = 2 pi i x (sum of residues of f at a1, ..., an)

Example 1

Evaluate contour Int_C $(z + 1)/(z^2 (z - 2))\ dz$, C: $|z| = 3$ (both poles inside). Res at $z = 2$ (simple): $(2 + 1)/4 = 3/4$. Res at $z = 0$ (order 2): $d/dz\ [(z+1)/(z - 2)]$ at $0 = [(z - 2) - (z + 1)]/(z - 2)^2 = -3/4$. Sum = 0, so integral = 0.

Example 2

contour Int_C dz/(z^2 + 4) over |z - i| = 2: pole z = 2i is inside (distance |2i - i| = 1 < 2), z = -2i is outside. Res at 2i = 1/(4i). Integral = 2 pi i (1/4i) = pi/2 .

Example 3

contour Int_C tan z dz over |z| = 2: poles of tan z at z = +/- pi/2 (|pi/2| = 1.57 < 2). Res of sin z/cos z at pi/2 = sin(pi/2)/(-sin(pi/2)) = -1. Same for -pi/2. Integral = 2 pi i (-2) = -4 pi i .

8. Definite Integrals Involving sin and cos

Type: Int (0 to 2 pi) F(cos theta, sin theta) d theta , F a rational function. Substitute z = e^(i theta) - theta from 0 to 2 pi means z goes once round the unit circle |z| = 1.

Substitution

- z = e^(i theta), dz = i z d theta, d theta = dz/(i z)
- cos theta = (z + 1/z)/2 , sin theta = (z - 1/z)/(2i)
- cos n theta = (z^n + z^-n)/2 ; sin n theta = (z^n - z^-n)/2i
- Integral = contour Int over |z| = 1 of G(z) dz = 2 pi i x Sum of residues inside |z| = 1

Example 1

Evaluate I = Int (0 to 2pi) d theta/(5 + 4 cos theta).

5 + 4 cos t = 5 + 2(z + 1/z) = (2z^2 + 5z + 2)/z = 2 (z + 2)(z + 1/2)/z. I = contour Int dz/(i z) x z/[2 (z + 2)(z + 1/2)] = (1/2i) contour Int dz/((z + 2)(z + 1/2)).

Poles: z = -2 (outside), z = -1/2 (inside). Res at -1/2 = 1/(-1/2 + 2) = 2/3. I = (1/2i)(2 pi i)(2/3) = 2 pi/3 .

Example 2

Evaluate Int (0 to 2pi) d theta/(a + b cos theta), a > b > 0. Standard result:

check: a = 5, b = 4 gives 2 pi/3 OK

Int (0 to 2 pi) d theta / (a + b cos theta) = 2 pi / sqrt(a^2 - b^2)

Example 3

Int (0 to 2pi) d theta/(2 + cos theta) = 2 pi / sqrt(4 - 1) = 2 pi/sqrt3 .

Example 4 - with cos 2 theta

I = Int (0 to 2pi) cos 2t /(5 - 4 cos t) dt. Take I = Re Int e^(2it)/(5 - 4 cos t) dt. With z = e^(it): 5 - 4cos t = (-2z^2 + 5z - 2)/z = -2(z - 2)(z - 1/2)/z. So Int = contour Int z^2 . z . dz /(i z . (-2)(z - 2)(z - 1/2)) = (-1/2i) contour Int z^2 dz/((z - 2)(z - 1/2)).

Only z = 1/2 inside: Res = (1/4)/(1/2 - 2) = -1/6. Integral = (-1/2i)(2 pi i)(-1/6) = pi/6 .

Quick revision - Unit II

- Int_C f dz = Int f(z(t)) z'(t) dt, circle: contour Int dz/(z - a) = 2 pi i
- Cauchy: analytic => integral = 0, formula: f(a) = (1/2 pi i) Int f/(z - a) dz
- Taylor: Sum f^(n)(a)(z - a)^n / n!, Laurent: analytic part + principal part
- Residue (simple pole) = lim (z - a) f(z); Residue theorem: Int = 2 pi i x Sum Res
- Trig integrals: z = e^(i theta), d theta = dz/(iz), poles inside |z| = 1

HANDWRITTEN NOTES

COMPLEX VARIABLES, PROBABILITY AND STATISTICS

Subject Code : 23A54403

UNIT - III

Probability Theory

Topics covered in this unit :

- Probability, probability axioms
- Addition law and multiplication law of probability
- Conditional probability, Bayes' theorem
- Random variables (discrete and continuous)
- Probability density functions and their properties
- Mathematical expectation

(Key terms are written in red , headings in purple/green, formulas on yellow.)

1. Basic Concepts of Probability

Term	Meaning
Random experiment	Outcome cannot be predicted with certainty, but all possible outcomes are known (tossing a coin)
Sample space S	Set of ALL possible outcomes (die: S = {1,2,3,4,5,6})
Event E	A subset of S (getting an even number: {2,4,6})
Simple / compound event	Single outcome / combination of outcomes
Sure and impossible event	S itself / empty set (null)
Mutually exclusive	Cannot occur together: A ∩ B = empty
Exhaustive events	Together they cover S: A1 ∪ A2 ∪ ... = S
Independent events	Occurrence of one does not affect the other
Equally likely	Each outcome has the same chance

1.1 Definitions of probability

Classical: P(E) = (number of favourable cases) / (total number of equally likely cases) = m / n

- Statistical (relative frequency): P(E) = lim (n -> inf) (m/n), where m = number of times E occurs in n trials.
- Axiomatic (Kolmogorov): see below.

1.2 Axioms of probability

Kolmogorov's axioms

- Axiom 1: 0 <= P(E) <= 1 for every event E (non-negativity)
- Axiom 2: P(S) = 1 (certainty)
- Axiom 3: If E1, E2, ... are mutually exclusive, then P(E1 ∪ E2 ∪ ...) = P(E1) + P(E2) + ... (additivity)

1.3 Consequences

Result	Formula
Impossible event	P(empty) = 0
Complement	P(A') = 1 - P(A)
Subset	If A is a subset of B then P(A) <= P(B)
Difference	P(A - B) = P(A) - P(A n B)
Neither A nor B	P(A' n B') = 1 - P(A U B)
Exactly one of A, B	P(A) + P(B) - 2 P(A n B)

Example 1

Two dice are thrown. Find P(sum = 8). Total outcomes = 36. Favourable: (2,6),(3,5),(4,4),(5,3),(6,2) = 5. P = 5/36 .

Example 2

A card is drawn from a pack of 52. P(King or heart)? Kings = 4, hearts = 13, King of hearts counted twice: (4 + 13 - 1)/52 = 16/52 = 4/13 .

Example 3

From a box with 4 red and 6 black balls, 3 are drawn. P(all red) = C(4,3)/C(10,3) = 4/120 = 1/30 . P(2 red, 1 black) = C(4,2) C(6,1)/120 = 36/120 = 3/10 .

2. Laws of Probability

2.1 Addition law

General (any two events)

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A \cup B) = P(A) + P(B) \text{ if } A, B \text{ are mutually exclusive}$$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(A \cap B) - P(B \cap C) - P(C \cap A) + P(A \cap B \cap C)$$

2.2 Conditional probability

Probability of A given that B has occurred :

$$P(A \mid B) = P(A \cap B) / P(B), \quad P(B) > 0$$

2.3 Multiplication law

General

$$P(A \cap B) = P(A)\, P(B \mid A) = P(B)\, P(A \mid B)$$

Independent events

$$P(A \cap B) = P(A)\, P(B) \text{ if } A, B \text{ are INDEPENDENT}$$

$$P(A \cap B \cap C) = P(A)\, P(B|A)\, P(C \mid A \cap B)$$

Independent vs mutually exclusive

- Mutually exclusive: $A \cap B$ = empty $\rightarrow P(A \cap B) = 0$ (cannot happen together)
- Independent: $P(A \cap B) = P(A)\, P(B)$
- Two events with non-zero probabilities that are mutually exclusive are never independent
- If A and B are independent, so are A and B', A' and B, A' and B'

Example 1

$P(A) = 0.5$, $P(B) = 0.4$, $P(A \cap B) = 0.2$. Then $P(A \cup B) = 0.5 + 0.4 - 0.2 = 0.7$.

$P(A|B) = 0.2/0.4 = 0.5 = P(A) \rightarrow$ A and B are independent.

Example 2 - drawing without replacement

Two cards are drawn without replacement from 52. P(both aces) = (4/52)(3/51) = 1/221 .

Example 3

A can solve a problem with probability 1/2, B with 1/3. If both try independently, P(problem solved) = 1 - P(neither) = 1 - (1/2)(2/3) = 2/3 .

Example 4

A bag has 5 white, 3 black balls. Two drawn one by one with no replacement. P(first white, second black) = (5/8)(3/7) = 15/56 .

3. Total Probability and Bayes' Theorem

3.1 Theorem of total probability

If B1, B2, ..., Bn are mutually exclusive and exhaustive events (a partition of S) with P(Bi) > 0, then for any event A:

$$P(A) = \sum_{i=1}^{n} P(B_i)\,P(A \mid B_i)$$

3.2 Bayes' theorem

Bayes' theorem

- $P(B_K \mid A) = P(B_K)\,P(A \mid B_K) \,/\, \sum_{i=1}^{n} P(B_i)\,P(A \mid B_i)$
- P(Bi) = prior probabilities (before the event), P(A | Bi) = likelihood ; P(BK | A) = posterior probability (after A is known)
- Used to find the probability of a cause given its effect

Tree method

Choose source Bi with P(Bi) → Event A occurs with P(A | Bi) → Add all branches for P(A); posterior = branch / total

Example 1 - Three machines

Machines M1, M2, M3 produce 50 %, 30 %, 20 % of items with defective rates 3 %, 4 %, 5 %. An item picked at random is defective. Find P(it came from M3).

Machine	P(Bi)	P(D | Bi)	Product
M1	0.50	0.03	0.015
M2	0.30	0.04	0.012
M3	0.20	0.05	0.010
Total	1.00		0.037

P(D) = 0.037. P(M3 | D) = 0.010/0.037 = 0.2703 (27 %).

Example 2 - Medical test

1 % of people have a disease. Test is 95 % positive for diseased and 2 % false positive for healthy. P(disease | positive) = (0.01 × 0.95)/(0.01 × 0.95 + 0.99 × 0.02) = 0.0095/0.0293 = 0.324 . Even a positive test gives only 32 % chance - because the disease is rare.

Example 3 - Urns

Urn I has 3 white, 2 black balls; Urn II has 2 white, 4 black. An urn is chosen at random and a ball is drawn - it is white. P(from Urn I) = (1/2)(3/5)/[(1/2)(3/5) + (1/2)(2/6)] = 0.3/(0.3 + 0.1667) = 0.643 .

4. Random Variables

A random variable (r.v.) X is a function that assigns a real number to each outcome of the sample space: $X : S \to R$. Written in capital letters (X), its values in small letters (x).

Discrete r.v.	Continuous r.v.
Takes countable values (finite or infinite list)	Takes any value in an interval (uncountable)
Number of heads, defects, customers	Height, weight, time, temperature

Discrete r.v.	Continuous r.v.
Described by probability mass function (p.m.f.) $p(x)$	Described by probability density function (p.d.f.) $f(x)$
$P(X = x) = p(x)$ can be > 0	$P(X = x) = 0$ for any single point

4.1 Discrete: probability mass function

p.m.f. conditions

- $p(x_i) = P(X = x_i) \geq 0$ for all i
- Sum of $p(x_i) = 1$
- Distribution function (c.d.f.): $F(x) = P(X \leq x)$ = Sum of $p(x_i)$ for $x_i \leq x$

Example

Two coins are tossed; X = number of heads. $S = \{HH, HT, TH, TT\}$.

x	0	1	2
$p(x)$	1/4	1/2	1/4
$F(x)$	1/4	3/4	1

Check: $1/4 + 1/2 + 1/4 = 1$. $P(X \geq 1) = 3/4$.

Example - find K

$p(x) = K\,x$ for $x = 1, 2, 3, 4, 5$. Sum $= K(1+2+3+4+5) = 15K = 1 \rightarrow K = 1/15$.

$P(X \leq 2) = 3/15 = 1/5$.

4.2 Continuous: probability density function

p.d.f. $f(x)$ properties

- $f(x) \geq 0$ for all x
- $\int_{-\infty}^{\infty} f(x)\,dx = 1$ (total area = 1)
- $P(a \leq X \leq b) = \int_a^b f(x)\,dx$ = area under the curve (end points do not matter)
- c.d.f.: $F(x) = P(X \leq x) = \int_{-\infty}^{x} f(t)\,dt$; $f(x) = F'(x)$
- F is non-decreasing, $F(-\infty) = 0$, $F(\infty) = 1$, $0 \leq F \leq 1$

Example 1

f(x) = K x^2 for 0 <= x <= 3, 0 elsewhere. Int (0 to 3) K x^2 dx = 9 K = 1 -> K = 1/9 . P(1 <= X <= 2) = (1/9)(8 - 1)/3 = 7/27 . F(x) = x^3/27 (0 <= x <= 3).

Example 2

f(x) = 2 e^(-2x), x >= 0 (exponential). P(X > 1) = Int (1 to inf) 2e^(-2x) dx = e^(-2) = 0.1353 . F(x) = 1 - e^(-2x).

Example 3

f(x) = K(1 - x^2), -1 <= x <= 1: Int = K [x - x^3/3] from -1 to 1 = K(4/3) = 1 -> K = 3/4 .

5. Mathematical Expectation

5.1 Definition

Mean = expected value = mu

Discrete: E(X) = Sum xi p(xi) Continuous: E(X) = Int (-inf to inf) x f(x) dx

Expectation of a function

E[g(X)] = Sum g(xi) p(xi) or Int g(x) f(x) dx

5.2 Variance and standard deviation

Var(X) = E[(X - mu)^2] = E(X^2) - [E(X)]^2 , sigma = sqrt(Var(X))

5.3 Properties of expectation and variance

Expectation	Variance
E(c) = c	Var(c) = 0
E(aX + b) = a E(X) + b	Var(aX + b) = a^2 Var(X)
E(X + Y) = E(X) + E(Y) (always)	Var(X + Y) = Var X + Var Y if independent
E(XY) = E(X)E(Y) if independent	Var(X - Y) = Var X + Var Y if independent

Expectation	Variance
E(X^2) >= [E(X)]^2	Var(X) >= 0

Moments

r-th moment about origin: mu'_r = E(X^r). r-th central moment: mu_r = E[(X - mu)^r]. mu_1 = 0, mu_2 = Var(X). Moment generating function: M(t) = E(e^(tX)); mu'_r = d^r M/dt^r at t = 0.

Example 1 - discrete

X = number of heads in 2 tosses: E(X) = 0(1/4) + 1(1/2) + 2(1/4) = 1 . E(X^2) = 0 + 1/2 + 4/4 = 3/2. Var(X) = 3/2 - 1 = 1/2 .

Example 2 - die

X = score on a die. E(X) = (1 + 2 + ... + 6)/6 = 3.5 . E(X^2) = 91/6. Var = 91/6 - 49/4 = 35/12 = 2.917 .

Example 3 - continuous

f(x) = x/2 for 0 <= x <= 2. Check: Int x/2 dx = 1 OK. E(X) = Int x (x/2) dx = [x^3/6] 0 to 2 = 4/3 . E(X^2) = Int x^3/2 dx = [x^4/8] = 2. Var = 2 - 16/9 = 2/9 .

Example 4 - gain in a game

You win Rs.100 if a die shows 6, lose Rs.20 otherwise. E(gain) = 100(1/6) - 20(5/6) = 16.67 - 16.67 = 0 -> fair game.

Example 5 - E(2X + 3)

If E(X) = 4 and Var(X) = 9: E(2X + 3) = 11; Var(2X + 3) = 4 x 9 = 36 .

6. More Results on Expectation

6.1 Moment generating function (M.G.F.)

$$M_X(t) = E(e^{tX}) = \sum e^{tx} p(x) \text{ or } \int e^{tx} f(x)\,dx$$

- Moments: mu'_r = (d^r / dt^r) M_X(t) at t = 0 ; so mean = M'(0), E(X^2) = M''(0).

- M.G.F. of aX + b is e^(bt) M_X(a t); for independent X, Y: M_(X+Y)(t) = M_X(t) M_Y(t).
- M.G.F. uniquely determines the distribution.

Example

p(x) = (1/2)^x, x = 1, 2, 3, ... Then M(t) = Sum (e^t/2)^x = (e^t/2)/(1 - e^t/2) = e^t/(2 - e^t). M'(0) = 2 -> E(X) = 2 .

6.2 Chebyshev's inequality

P(|X - mu| >= K sigma) <= 1/K^2 or P(|X - mu| < K sigma) >= 1 - 1/K^2

Valid for any distribution with finite mean and variance. Example: for K = 2, at least 75 % of values lie within mu +/- 2 sigma; for K = 3, at least 88.9 %.

Example

A r.v. has mean 12 and variance 9. P(6 < X < 18) >= ? Here K sigma = 6, sigma = 3 -> K = 2. Bound = 1 - 1/4 = 0.75 .

6.3 Median, mode (continuous)

- Median M: Int (-inf to M) f(x) dx = 1/2. Mode: value where f(x) is maximum (f' = 0, f'' < 0).
- Example: f(x) = 3x^2 on (0,1): median: M^3 = 1/2 -> M = 0.794; mode = 1 (f increasing).

7. Practice - Worked Problems

Problem 1 - Conditional probability

In a class 60 % are boys, 30 % of boys and 20 % of girls play cricket. A student plays cricket. P(boy)? P(play) = 0.6(0.3) + 0.4(0.2) = 0.18 + 0.08 = 0.26. P(boy | play) = 0.18/0.26 = 0.692 .

Problem 2 - at least one

A coin is tossed 4 times. P(at least one head) = 1 - (1/2)^4 = 15/16 .

Problem 3 - Independent events

P(A) = 1/3, P(B) = 1/4, independent. P(A ∪ B) = 1/3 + 1/4 - 1/12 = 1/2 . P(A' ∩ B') = (2/3)(3/4) = 1/2 (= 1 - P(A ∪ B)).

Problem 4 - Bayes with three boxes

Box 1: 2 red 3 blue, Box 2: 3 red 2 blue, Box 3: 4 red 1 blue. A box is chosen randomly and a red ball is drawn. P(Box 3)? P(R) = (1/3)(2/5 + 3/5 + 4/5) = 3/5. P(B3 | R) = (1/3)(4/5)/(3/5) = 4/9 .

Problem 5 - Spam filter

30 % of mails are spam. The word 'offer' appears in 80 % of spam and 10 % of non-spam. P(spam | offer) = 0.3 x 0.8/(0.3 x 0.8 + 0.7 x 0.1) = 0.24/0.31 = 0.774 .

Problem 6 - Discrete distribution

x	-2	-1	0	1	2
p(x)	0.1	0.2	0.3	0.3	0.1

E(X) = -0.2 - 0.2 + 0 + 0.3 + 0.2 = 0.1 . E(X^2) = 0.4 + 0.2 + 0 + 0.3 + 0.4 = 1.3. Var = 1.3 - 0.01 = 1.29 .

Problem 7 - Continuous with c.d.f.

f(x) = 6x(1 - x), 0 < x < 1. Check: 6[x^2/2 - x^3/3] = 1. E(X) = ∫ 6x^2(1 - x) dx = 6(1/3 - 1/4) = 1/2 . E(X^2) = 6(1/4 - 1/5) = 3/10. Var = 0.3 - 0.25 = 0.05 .

Problem 8 - Find K and probability

f(x) = K e^(-3x), x > 0. ∫ = K/3 = 1 -> K = 3. P(X < 1) = 1 - e^(-3) = 0.9502 . Mean = 1/3.

Problem 9 - Counting and probability

A committee of 3 is formed from 5 men and 4 women. P(exactly 2 women) = C(4,2) C(5,1)/C(9,3) = 30/84 = 5/14 .

Problem 10 - Birthday-type

P(all 3 people have different birthdays) = (365/365)(364/365)(363/365) = 0.9918 ; so P(at least two share) = 0.0082.

Key formula	Use
nPr = n!/(n-r)!, nCr = n!/(r!(n-r)!)	Counting favourable cases
P(A|B) = P(A∩B)/P(B)	Conditional probability
Total probability + Bayes	Cause and effect problems
E(X), Var(X), M(t)	Mean, spread, moments

Quick revision - Unit III

- Axioms: 0 <= P <= 1, P(S) = 1, additivity; P(A ∪ B) = P(A) + P(B) - P(A ∩ B)
- P(A|B) = P(A ∩ B)/P(B); independent: P(A ∩ B) = P(A)P(B)
- Total probability P(A) = Sum P(Bi)P(A|Bi); Bayes: P(Bk|A) = P(Bk)P(A|Bk)/P(A)
- p.m.f.: Sum p = 1; p.d.f.: Int f dx = 1, f >= 0, f = F'
- E(X) = Sum x p or Int x f dx; Var = E(X^2) - [E(X)]^2

HANDWRITTEN NOTES

COMPLEX VARIABLES, PROBABILITY AND STATISTICS

Subject Code : 23A54403

UNIT - IV

Binomial, Poisson and Normal Distributions

Topics covered in this unit :

- Probability distributions: Binomial distribution
- Poisson distribution; Poisson approximation to the binomial
- Normal distribution and its properties
- Mean, variance, M.G.F., applications and numerical problems

(Key terms are written in red , headings in purple/green, formulas on yellow.)

1. Probability Distribution - Overview

A probability distribution gives the probabilities of all values of a random variable. Discrete: Binomial, Poisson (p.m.f.). Continuous: Normal, uniform, exponential (p.d.f.).

Distribution	Type	Parameters	Mean	Variance
Binomial B(n, p)	Discrete	n, p	n p	n p q
Poisson P(lambda)	Discrete	lambda	lambda	lambda
Normal N(mu, sigma^2)	Continuous	mu, sigma	mu	sigma^2
Uniform U(a, b)	Continuous	a, b	(a+b)/2	(b-a)^2/12
Exponential	Continuous	theta	1/theta	1/theta^2

2. Binomial Distribution

2.1 Bernoulli trials

A trial with only two outcomes - success (probability p) and failure (probability q = 1 - p). Repeated n independent trials with the same p -> binomial setting.

Conditions for binomial distribution

- Fixed number n of trials
- Each trial has two outcomes: success / failure
- Trials are independent
- Probability of success p is constant

2.2 p.m.f.

$$P(X = x) = C(n, x)\, p^x\, q^{(n - x)}, \quad x = 0, 1, 2, \ldots, n$$

Notation: $X \sim B(n, p)$. Sum of probabilities $= (q + p)^n = 1$. (Binomial

expansion of (q + p)^n, hence the name.)

2.3 Constants

Quantity	Value
Mean	mu = n p
Variance	sigma^2 = n p q (variance < mean)
Standard deviation	sqrt(n p q)
Mode	integer part of (n + 1) p
Recurrence	P(x + 1) = [(n - x)/(x + 1)] (p/q) P(x)
M.G.F.	M(t) = (q + p e^t)^n
Skewness / Kurtosis	(q - p)/sqrt(npq) ; 3 + (1 - 6pq)/(npq)

2.4 Derivation of mean and variance

E(X) = Sum x C(n,x) p^x q^(n-x) = n p Sum C(n-1, x-1) p^(x-1) q^(n-x) = n p (q + p)^(n-1) = n p . E[X(X-1)] = n(n-1)p^2 -> Var = n(n-1)p^2 + np - n^2p^2 = n p q .

Example 1

A fair coin is tossed 6 times. Find P(exactly 4 heads), P(at least 4 heads).

n = 6, p = 1/2: P(4) = C(6,4)(1/2)^6 = 15/64 = 0.2344 . P(X >= 4) = (15 + 6 + 1)/64 = 22/64 = 0.34375 .

Example 2

The probability that a bulb is defective is 0.1. In a sample of 10, P(no defective) = (0.9)^10 = 0.3487 ; P(at most 2 defective) = 0.3487 + 10(0.1)(0.9)^9 + 45(0.01)(0.9)^8 = 0.3487 + 0.3874 + 0.1937 = 0.9298 .

Example 3 - mean and variance given

Mean of a binomial = 4 and variance = 3. Then n p = 4, n p q = 3 -> q = 3/4, p = 1/4, n = 16. P(X = 2) = C(16,2)(1/4)^2(3/4)^14 = 120 x 0.0625 x 0.01782 = 0.1336 .

Example 4 – expected frequencies

Out of 800 families with 5 children each, how many are expected to have 3 boys? $p = 1/2$: $P(3) = C(5,3)/32 = 10/32$. Frequency $= 800 \times 10/32 = 250$ families.

Example 5

A multiple-choice test has 5 questions, each with 4 options. If answered by guessing, P(at least 4 correct) $= C(5,4)(1/4)^4(3/4) + (1/4)^5 = 15/1024 + 1/1024 = 16/1024 = 0.0156$.

3. Poisson Distribution

Introduced by S. D. Poisson (1837). Used for rare events in a fixed interval of time / space: number of accidents per day, telephone calls per minute, misprints per page, defects per metre of cloth, radioactive decays.

3.1 p.m.f.

$$P(X = x) = e^{-\lambda}\lambda^x / x!, \quad x = 0, 1, 2, \ldots$$
$$(\lambda > 0)$$

Notation: $X \sim P(\lambda)$, where λ = average number of occurrences.

Quantity	Value
Mean	λ
Variance	λ (mean = variance)
Standard deviation	$\sqrt{\lambda}$
Mode	integer part of λ (if λ is an integer: λ and $\lambda - 1$)
Recurrence	$P(x+1) = [\lambda/(x+1)]\,P(x)$
M.G.F.	$M(t) = \exp[\lambda(e^t - 1)]$
Additive	$X_1 \sim P(\lambda_1)$, $X_2 \sim P(\lambda_2)$ independent $\Rightarrow X_1 + X_2 \sim P(\lambda_1 + \lambda_2)$

3.2 Poisson as a limit of the binomial

Poisson approximation to the Binomial

- If n is large (n -> infinity), p is small (p -> 0), and n p = lambda stays finite, then
- C(n,x) p^x q^(n-x) -> e^(-lambda) lambda^x / x!
- Rule of thumb: n >= 20 and p <= 0.05 (or n p < 5)
- Proof idea: q^(n-x) = (1 - lambda/n)^(n-x) -> e^(-lambda), and n(n-1)...(n-x+1)/n^x -> 1

Derivation (short)

P(x) = [n(n-1)...(n-x+1)/x!] (lambda/n)^x (1 - lambda/n)^(n-x) = (lambda^x/x!) [1(1 - 1/n)...(1 - (x-1)/n)] (1 - lambda/n)^n (1 - lambda/n)^(-x). As n -> inf: product -> 1, (1 - lambda/n)^n -> e^(-lambda), last factor -> 1. Hence P(x) -> e^(-lambda) lambda^x / x! .

Example 1

A shop receives on average 3 calls per hour. P(exactly 2 calls in an hour) = e^-3 3^2/2! = 0.0498 x 4.5 = 0.2240 . P(no call) = e^-3 = 0.0498 . P(at least 1) = 0.9502 .

Example 2 - approximation

1 % of items are defective, in a sample of 200 find P(exactly 3 defective). n = 200, p = 0.01, lambda = 2. P(3) = e^-2 2^3/3! = 0.1353 x 8/6 = 0.1804 .

Example 3

A book of 500 pages has 500 misprints distributed randomly. Average = 1 per page. P(page has at least 2 misprints) = 1 - P(0) - P(1) = 1 - e^-1(1 + 1) = 1 - 0.7358 = 0.2642 . In a book of 1000 such pages: 264 pages expected.

Example 4 - mean and variance

If X ~ Poisson and P(X = 1) = P(X = 2): lambda e^-l = l^2 e^-l/2 -> lambda = 2 . Then P(X = 4) = e^-2 2^4/4! = 0.1353 x 16/24 = 0.0902 ; mean = variance = 2.

Example 5 - cars

The number of accidents on a road per week is Poisson with mean 2. P(at most 2 accidents in a week) = e^-2 (1 + 2 + 2) = 0.1353 x 5 = 0.6767 .

Basis	Binomial	Poisson
Trials	Fixed n	Unlimited (rate based)
Parameters	n, p	lambda only
Mean vs variance	Variance < mean	Mean = variance
Possible values	0 to n	0 to infinity

4. Normal Distribution

The most important continuous distribution - Gaussian , introduced by De Moivre and developed by Gauss, Laplace . Many natural variables (heights, marks, errors in measurement) are normal. By the central limit theorem sums / means of many independent r.v.s are approximately normal.

4.1 p.d.f.

$$f(x) = [\ 1 / (\sigma \sqrt{2\pi})\ ] \exp[\ -(x-\mu)^2 / (2\sigma^2)\ ],\ -\infty < x < \infty$$

X ~ N(mu, sigma^2). Parameters: mean mu and standard deviation sigma (sigma > 0).

4.2 Standard normal variate

$$Z = (X-\mu)/\sigma \sim N(0,1)\quad \phi(z) = (1/\sqrt{2\pi})\, e^{(-z^2/2)}$$

Tables give the area Phi(z) = P(Z <= z) or area from 0 to z. Use symmetry: P(Z <= -z) = 1 - P(Z <= z).

4.3 Properties of the normal curve

Properties

- Bell shaped, symmetric about x = mu; mean = median = mode = mu
- Total area under curve = 1; area on each side of mu = 0.5
- Points of inflection at x = mu +/- sigma; asymptotic to x-axis (never touches)
- Skewness = 0; Kurtosis = 3 (mesokurtic); mean deviation = 0.7979 sigma; quartile deviation = 0.6745 sigma
- M.G.F. = exp(mu t + sigma^2 t^2 / 2); central moments: mu_2 = sigma^2, odd moments = 0
- Sum of independent normals is normal: N(m1, s1^2) + N(m2, s2^2) = N(m1 + m2, s1^2 + s2^2)
- Maximum ordinate = 1/(sigma sqrt(2 pi)) at x = mu

Empirical rule (areas)

Interval	Area (probability)
mu +/- 1 sigma	0.6826 (68.26 %)
mu +/- 2 sigma	0.9544 (95.44 %)
mu +/- 3 sigma	0.9973 (99.73 %)
mu +/- 1.96 sigma	0.95 ; mu +/- 2.58 sigma: 0.99

Standard normal table values to remember

z	0.5	1.0	1.5	1.96	2.0	2.58	3.0
P(0<Z<z)	0.1915	0.3413	0.4332	0.4750	0.4772	0.4951	0.4987

Example 1

X ~ N(50, 10^2). P(45 < X < 62)? z1 = (45 - 50)/10 = -0.5, z2 = (62 - 50)/10 = 1.2. P = P(0<Z<0.5) + P(0<Z<1.2) = 0.1915 + 0.3849 = 0.5764 .

Example 2

Marks are normal with mean 60, sigma 10. P(X > 75): z = 1.5 -> 0.5 - 0.4332 = 0.0668 . If 500 students, about 33 score above 75. Top 10 % cut-off: z = 1.28 -> X = 60 + 12.8 = 72.8 .

Example 3 - finding mu and sigma

In a normal distribution 31 % of items are below 45 and 8 % are above 64. P(Z < -0.495) = 0.31 -> (45 - mu)/sigma = -0.495, P(Z > 1.405) = 0.08 -> (64 - mu)/sigma = 1.405. Subtract: 19/sigma = 1.9 -> sigma = 10 , mu = 45 + 4.95 = 49.95 (~50) .

Example 4 - bulbs

Life of bulbs ~ N(1200, 200^2) hours. P(X < 900) = P(Z < -1.5) = 0.0668. In 1000 bulbs, 67 expected to fail before 900 hrs. P(1000 < X < 1400) = 2 x 0.3413 = 0.6826 .

4.4 Normal approximation to binomial

For large n (n p > 5 and n q > 5): B(n, p) ~ N(n p, n p q). Apply continuity correction : P(X = K) ~ P(K - 1/2 < X < K + 1/2).

Example 5

A fair coin is tossed 100 times. P(48 to 55 heads)? mu = 50, sigma = 5. Continuity: 47.5 to 55.5 -> z = -0.5 to 1.1: 0.1915 + 0.3643 = 0.5558 .

Basis	Binomial	Poisson	Normal
Variable	Discrete	Discrete	Continuous
Shape	Skew unless p = 1/2	Skew, positive	Symmetric bell
Parameters	n, p	lambda	mu, sigma
Limit of	-	B(n,p), n large, p small	B(n,p), n large, p moderate

5. Other Useful Distributions (brief)

Uniform U(a, b)

f(x) = 1/(b - a) on a <= x <= b. Mean (a + b)/2, variance (b - a)^2/12.

Example: waiting time for a bus arriving uniformly every 10 min: P(wait > 7) = 3/10.

Exponential

f(x) = theta e^(-theta x), x > 0; mean 1/theta, variance 1/theta^2; memoryless. Time between Poisson events: if calls come at rate lambda per hour, the waiting time is exponential with theta = lambda.

Geometric

P(X = x) = q^(x-1) p, x = 1, 2, ... (trials until first success), mean 1/p, variance q/p^2.

Hypergeometric

Sampling without replacement: P(X = x) = C(M, x) C(N - M, n - x)/C(N, n). Becomes binomial for large N.

6. Derivations and Fitting

6.1 Binomial M.G.F. and moments

M(t) = Sum e^(tx) C(n,x) p^x q^(n-x) = Sum C(n,x)(p e^t)^x q^(n-x) = (q + p e^t)^n. Differentiate: M'(0) = n p (mean). M''(0) = n(n-1)p^2 + n p, so Var = n p - n p^2 = n p q.

6.2 Poisson mean and variance

E(X) = Sum x e^-l l^x/x! = l e^-l Sum l^(x-1)/(x-1)! = lambda. E[X(X-1)] = l^2 -> E(X^2) = l^2 + l -> Var = lambda.

6.3 Normal: area check

Put z = (x - mu)/sigma: Int f(x) dx = (1/sqrt(2 pi)) Int e^(-z^2/2) dz = (1/sqrt(2 pi)) sqrt(2 pi) = 1 (using Int e^(-a z^2) dz = sqrt(pi/a)).

6.4 Steps for a normal-distribution problem

Write X ~ N(mu, sigma^2) → Convert limits to z = (x − mu)/sigma → Sketch the curve and shade area

Read table values using symmetry → Add / subtract areas, multiply by N if frequency asked

Required	Use
P(Z < a), a > 0	0.5 + A(a)
P(Z > a), a > 0	0.5 − A(a)
P(−a < Z < b)	A(a) + A(b)
P(a < Z < b), 0 < a < b	A(b) − A(a)
P(Z < −a)	0.5 − A(a)

(A(z) = area from 0 to z in the table.)

6.5 Fitting a binomial distribution

Given observed data x with frequency f: (1) find mean = Sum f x / Sum f; (2) mean = n p → p = mean/n; (3) expected frequency = N × C(n,x) p^x q^(n−x).

Example - fit

5 coins tossed 320 times; mean number of heads found = 2.5 → p = 0.5. Expected frequencies: N P(x) = 320 × C(5,x)/32 = 10, 50, 100, 100, 50, 10 for x = 0 … 5.

6.6 Fitting a Poisson distribution

Mean lambda = Sum f x / Sum f. Expected frequency = N e^−l l^x/x!. Use recurrence: P(x + 1) = l/(x + 1) P(x).

x (deaths)	0	1	2	3	4
Observed f	109	65	22	3	1

N = 200; Sum f x = 0 + 65 + 44 + 9 + 4 = 122 → lambda = 0.61 . P(0) =

e^-0.61 = 0.5434 -> expected 108.7; P(1) = 0.3315 -> 66.3; P(2) = 0.1011 -> 20.2; P(3) = 0.0206 -> 4.1; P(4) = 0.0031 -> 0.6. Good fit with observed values.

6.7 More problems

(a) A machine makes 5 % defective parts. In a box of 20, P(exactly 1 defective) = C(20,1)(0.05)(0.95)^19 = 0.3774 (Binomial); Poisson approx with lambda = 1 gives e^-1 = 0.3679.

(b) Heights ~ N(165, 6^2) cm. P(height > 177) = P(Z > 2) = 0.0228 ; P(159 < X < 171) = 0.6826 .

(c) In 1 minute a switchboard gets calls Poisson(lambda = 2). P(more than 3 calls) = 1 - e^-2(1 + 2 + 2 + 1.333) = 1 - 0.8571 = 0.1429 .

(d) Weights of 1000 students ~ N(60, 8^2) Kg: number between 52 and 68 = 1000 x 0.6826 = 683 .

7. Practice Problems with Answers

Q	Problem	Answer
1	Binomial n = 8, p = 0.5: P(X = 5)	56/256 = 0.2188
2	Binomial n = 5, p = 1/3: mean and variance	5/3 ; 10/9
3	Poisson lambda = 4: P(X <= 1)	e^-4 (1 + 4) = 0.0916
4	Poisson lambda = 0.5: P(X = 0)	e^-0.5 = 0.6065
5	Z: P(-1.2 < Z < 0.8)	0.3849 + 0.2881 = 0.6730
6	N(100, 15^2): P(X > 130)	P(Z > 2) = 0.0228
7	Defect 2 %, sample 100: P(0 defects), Poisson	e^-2 = 0.1353
8	Binomial: 10 trials, p = 0.2, P(X >= 1)	1 - 0.8^10 = 0.8926

Quick revision - Unit IV

- Binomial: P(x) = C(n,x) p^x q^(n-x); mean np, variance npq
- Poisson: P(x) = e^-l l^x/x!; mean = variance = lambda; limit of binomial (n large, p small, np = lambda)
- Normal: bell shaped, mean = median = mode; Z = (X - mu)/sigma; 68-95-99.7 rule
- Normal approximation to binomial with continuity correction

HANDWRITTEN NOTES

COMPLEX VARIABLES, PROBABILITY AND STATISTICS

Subject Code : 23A54403

UNIT - V

Estimation, Large Sample Tests and Confidence Intervals

Topics covered in this unit :

- Estimation: parameters, statistics, sampling distribution
- Point estimation
- Null and alternative hypothesis; critical and acceptance regions
- Level of significance, two types of errors, power of the test
- Large sample tests: single proportion, difference of proportions
- Large sample tests: single mean, difference of means
- Confidence intervals: one-sample and two-sample problems

(Key terms are written in red , headings in purple/green, formulas on yellow.)

1. Population, Sample, Parameter, Statistic

Term	Meaning
Population	Complete collection of items under study (size N)
Sample	Part of the population actually observed (size n)
Random sampling	Every unit has equal chance of selection (with / without replacement)
Parameter	Numerical characteristic of the population (mu, sigma, P) - constants
Statistic	Function of sample values (x-bar, s, p) - random variables
Large sample	n >= 30; small sample: n < 30

	Population (parameter)	Sample (statistic)
Mean	mu	x-bar
Standard deviation	sigma	s
Proportion	P	p
Size	N	n

1.1 Sampling distribution

The probability distribution of a statistic over all possible samples of size n is its sampling distribution . Standard error (S.E.) = standard deviation of the sampling distribution of a statistic.

Statistic	Mean	Standard error
Sample mean x-bar	mu	sigma / sqrt(n)
Sample proportion p	P	sqrt(P Q / n)
Difference of means x1 - x2	mu1 - mu2	sqrt(s1^2/n1 + s2^2/n2)
Difference of proportions p1 - p2	P1 - P2	sqrt(P1Q1/n1 + P2Q2/n2)

Central Limit Theorem (C.L.T.)

- If x1, ..., xn is a random sample of size n >= 30 from any population with mean mu and variance sigma^2, then x-bar is approximately N(mu, sigma^2/n)
- For a normal population, x-bar is exactly normal for any n
- Basis of all large sample tests : statistic ~ N(0,1) after standardising

Finite population correction: S.E. is multiplied by sqrt((N - n)/(N - 1)) when sampling without replacement from small N.

2. Point Estimation

Estimation = using sample data to find the value of an unknown population parameter. Point estimate = a single value (e.g. x-bar for mu). Interval estimate = a range (confidence interval).

2.1 Characteristics of a good estimator

Property	Meaning
Unbiasedness	E(T) = theta. e.g. E(x-bar) = mu, E(s^2) = sigma^2 (with n - 1 divisor)
Consistency	T -> theta as n -> infinity (stochastically)
Efficiency	Among unbiased estimators, the one with least variance
Sufficiency	Contains all the information about theta present in the sample

2.2 Methods of point estimation

Method	Idea
Method of moments (Pearson)	Equate sample moments to population moments and solve
Maximum likelihood (Fisher)	Choose theta that maximises L(theta) = Product f(xi; theta)

Method	Idea
Least squares	Minimise sum of squared deviations

2.3 Common point estimates

Parameter	Estimator
Population mean mu	x-bar = Sum x / n
Population variance sigma^2	s^2 = Sum (x - x-bar)^2 / (n - 1) (unbiased)
Population proportion P	p = x / n
Poisson lambda	x-bar
Exponential theta	1 / x-bar

Example (MLE)

Poisson: L = Product e^-l l^xi / xi!. log L = -n l + (Sum xi) log l - Sum log xi!. d/dl = -n + Sum xi / l = 0 -> lambda-hat = x-bar .

3. Testing of Hypothesis

3.1 Hypotheses

Term	Meaning
Statistical hypothesis	Statement/assumption about a population parameter or distribution
Null hypothesis H0	Hypothesis of no difference / to be tested, has the equality sign (e.g. H0: mu = mu0)
Alternative hypothesis H1	Hypothesis accepted if H0 is rejected (mu != mu0, mu > mu0, mu < mu0)
Simple / composite	Specifies parameter fully / does not

- Two-tailed test: H1: mu != mu0 (difference in either direction).
- One-tailed test: H1: mu > mu0 (right tailed) or H1: mu < mu0 (left tailed).

3.2 Test statistic, critical and acceptance regions

Test statistic = quantity computed from the sample (for large samples $Z = (\text{statistic} - E(\text{statistic}))/S.E. \sim N(0,1)$). The set of values of the statistic for which H0 is rejected is the critical (rejection) region ; the remaining values form the acceptance region . The boundary values are critical values .

3.3 Level of significance

alpha = P(reject H0 when H0 is true) ; usually 5 % or 1 % . It fixes the size of the critical region.

alpha	Two-tailed	Right-tailed	Left-tailed
1 % (0.01)	$\|Z\| > 2.58$	$Z > 2.33$	$Z < -2.33$
5 % (0.05)	$\|Z\| > 1.96$	$Z > 1.645$	$Z < -1.645$
10 % (0.10)	$\|Z\| > 1.645$	$Z > 1.28$	$Z < -1.28$

3.4 Two types of errors and power

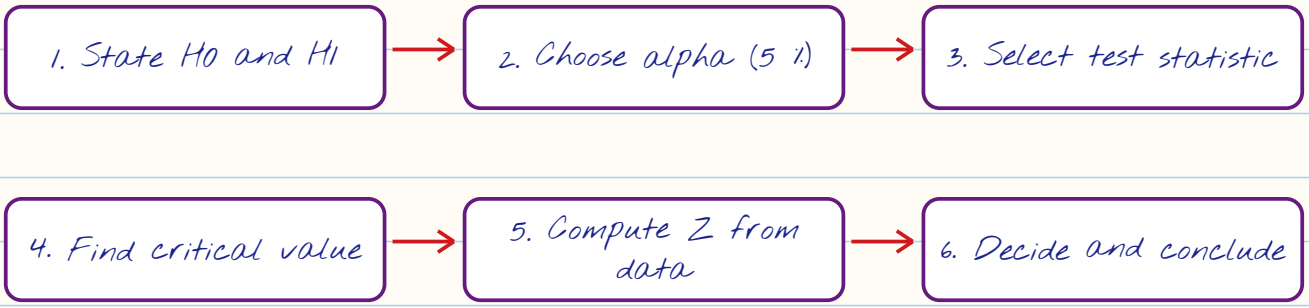
	H0 true	H0 false
Reject H0	Type I error (alpha) - producer's risk	Correct decision (power)
Accept H0	Correct decision	Type II error (beta) - consumer's risk

alpha = P(Type I error) = P(reject H0 | H0 true) beta = P(Type II error) = P(accept H0 | H0 false)

Power of the test = 1 - beta = P(reject H0 | H1 true)

For fixed n, reducing alpha increases beta . Increasing sample size reduces both. A good test has high power .

3.5 General procedure



Decision: if |Z calculated| > Z critical (table) -> reject H0 (significant); else accept / do not reject H0 .

4. Large Sample Tests

4.1 Test for single mean

H0: mu = mu0. Sample size n >= 30, mean x-bar:

Z = (x-bar - mu0) / (sigma / sqrt(n)) (use s if sigma unknown)

Example 1

A sample of 100 bulbs has mean life 1570 hrs, S.D. 120 hrs. Test at 5 % whether population mean = 1600 hrs. H0: mu = 1600; H1: mu != 1600. Z = (1570 - 1600)/(120/10) = -30/12 = -2.5 . |Z| = 2.5 > 1.96 -> reject H0 : the mean life differs from 1600 hrs.

Example 2 - one-tailed

A company claims average pack weight 500 g. 64 packs have mean 495 g, S.D. 16 g. Is weight less than claimed (5 %)? H1: mu < 500. Z = (495 - 500)/(16/8) = -2.5 < -1.645 -> reject H0, weight is significantly less.

4.2 Test for difference of means

H0: mu1 = mu2 (two independent large samples):

Z = (x1-bar - x2-bar) / sqrt(s1^2/n1 + s2^2/n2)

If sigma1 = sigma2 = sigma (common): S.E. = sigma sqrt(1/n1 + 1/n2).

Example 3

Two groups: n1 = 50, mean 82, S.D. 10; n2 = 60, mean 78, S.D. 12. Test equality of means at 5 %. S.E. = sqrt(100/50 + 144/60) = sqrt(2 + 2.4) = 2.098. Z = 4/2.098 = 1.907 < 1.96 -> accept H0 (difference not significant).

Example 4

Mean wages in two cities: 400 workers earn mean Rs.1500, S.D. 100 (city A); 500 workers earn Rs.1470, S.D. 120 (city B). S.E. = sqrt(10000/400 + 14400/500) = sqrt(25 + 28.8) = 7.335. Z = 30/7.335 = 4.09 > 2.58 -> highly significant, reject H0.

4.3 Test for single proportion

H0: P = P0. p = x/n = sample proportion, Q0 = 1 - P0.

$$Z = (p - P0) / \sqrt{P0\ Q0 / n}$$

Example 5

In 400 tosses of a coin 216 heads occur. Is the coin fair at 5 %? p = 216/400 = 0.54, P0 = 0.5. S.E. = sqrt(0.25/400) = 0.025. Z = 0.04/0.025 = 1.6 < 1.96 -> accept H0 : coin is fair.

Example 6

A manufacturer claims at most 5 % defective. In a sample of 200, 16 are defective. H0: P = 0.05; H1: P > 0.05. p = 0.08, S.E. = sqrt(0.05 x 0.95/200) = 0.01541. Z = 0.03/0.01541 = 1.947 > 1.645 -> reject H0 at 5 % (but accept at 1 %).

4.4 Test for difference of proportions

H0: P1 = P2. Pooled proportion P-hat = (n1 p1 + n2 p2)/(n1 + n2) = (x1 + x2)/(n1 + n2).

$$Z = (p1 - p2) / \sqrt{P\text{-}hat\ Q\text{-}hat\ (1/n1 + 1/n2)}$$

Example 7

In sample A, 20 of 200 are smokers, in sample B, 18 of 100. Test equality at 5 %. p1 = 0.10, p2 = 0.18, P-hat = 38/300 = 0.1267, Q-hat = 0.8733. S.E. = sqrt(0.1267 x 0.8733 (1/200 + 1/100)) = sqrt(0.11064 x 0.015) = 0.0407. Z = -0.08/0.0407 = -1.96 ; |Z| = 1.96 ~ critical -> borderline, reject H0 at 5 % (marginally).

Example 8

A machine A produces 40 defective out of 500, machine B 30 out of 400. P-hat = 70/900 = 0.0778; S.E. = sqrt(0.0778 x 0.9222 x (1/500 + 1/400)) = sqrt(0.07174 x 0.0045) = 0.01797. p1 = 0.08, p2 = 0.075: Z = 0.005/0.01797 = 0.278 -> accept H0; no difference between machines.

Test	H0	Test statistic Z
Single mean	mu = mu0	(x-bar - mu0)/(sigma/sqrt n)
Two means	mu1 = mu2	(x1 - x2)/sqrt(s1^2/n1 + s2^2/n2)
Single proportion	P = P0	(p - P0)/sqrt(P0Q0/n)
Two proportions	P1 = P2	(p1 - p2)/sqrt(PQ(1/n1 + 1/n2))

5. Confidence Intervals

A confidence interval (C.I.) is a range, computed from sample data, that contains the true parameter with a stated probability (confidence level 1 - alpha , e.g. 95 %). Values: 90 % -> 1.645; 95 % -> 1.96; 99 % -> 2.58 .

5.1 One-sample problems

Parameter	(1 - alpha) 100 % confidence interval
Mean mu (sigma Known / n large)	x-bar +/- Z(alpha/2) sigma/sqrt(n)
Mean mu (sigma unKnown, n >= 30)	x-bar +/- Z(alpha/2) s/sqrt(n)
Proportion P	p +/- Z(alpha/2) sqrt(p q / n)

Example 1

A sample of 100 has mean 52, S.D. 8. 95 % C.I. for mu: 52 +/- 1.96 (8/10) = 52 +/- 1.568 = (50.43, 53.57) . 99 %: 52 +/- 2.58(0.8) = (49.94, 54.06) .

Example 2 - proportion

In a survey of 400 voters 220 support a candidate. p = 0.55; S.E. = sqrt(0.55 x 0.45/400) = 0.0249. 95 % C.I. = 0.55 +/- 1.96(0.0249) = (0.501, 0.599) .

Example 3 - sample size

To estimate mean with margin of error E = 2 at 95 %, sigma = 10: n = (Z sigma/E)^2 = (1.96 x 10/2)^2 = 9.8^2 = 96.04 -> 97 .

5.2 Two-sample problems

Parameter	Confidence interval
mu1 - mu2	(x1 - x2) +/- Z(alpha/2) sqrt(s1^2/n1 + s2^2/n2)
P1 - P2	(p1 - p2) +/- Z(alpha/2) sqrt(p1q1/n1 + p2q2/n2)

Example 4

n1 = 100, x1 = 75, s1 = 10; n2 = 120, x2 = 70, s2 = 12. S.E. = sqrt(1 + 1.2) = 1.483. 95 % C.I. for mu1 - mu2 = 5 +/- 1.96(1.483) = 5 +/- 2.907 = (2.09, 7.91) . Does not contain 0 -> the means differ significantly.

Example 5 - proportions

p1 = 60/200 = 0.30, p2 = 40/150 = 0.267. S.E. = sqrt(0.3 x 0.7/200 + 0.267 x 0.733/150) = sqrt(0.00105 + 0.0013) = 0.0485. 95 % C.I. = 0.033 +/- 0.095 = (-0.062, 0.128) -> contains 0 -> no significant difference.

Relation between test and C.I.

- Two-tailed test at level alpha rejects H0: mu = mu0 iff mu0 lies outside the (1 - alpha) C.I.
- Wider interval = more confidence but less precision, larger n -> narrower interval
- Margin of error E = Z(alpha/2) x S.E.

6. Additional Concepts and Worked Problems

6.1 p-value approach

p-value = probability (under H0) of getting a result at least as extreme as the one observed. Reject H0 if p-value < alpha. For a two-tailed Z test: p = 2 P(Z > |z|).

Example: Z = 2.5 -> p = 2(0.0062) = 0.0124 < 0.05 -> reject H0 at 5 % but accept at 1 %.

6.2 Sampling distribution example

Population mean 80, S.D. 12. Sample of 36: S.E. = 12/6 = 2. P(x-bar > 83) = P(Z > 1.5) = 0.0668 . P(77 < x-bar < 83) = 2 x 0.4332 = 0.8664 .

6.3 Power of a test - example

H0: mu = 50 vs H1: mu = 52, sigma = 6, n = 36 (S.E. = 1), reject if x-bar > 51.645 (alpha = 5 %, one-tailed). If true mu = 52: beta = P(x-bar < 51.645 | mu = 52) = P(Z < -0.355) = 0.361. Power = 1 - 0.361 = 0.639 .

alpha	beta	Power
decreases	increases	decreases
n increases	decreases	increases

6.4 Steps - which test to use?

Is it a mean or a proportion? → One sample or two samples? → Is n >= 30? (large sample Z test)

Is H1 one-tailed or two-tailed? → Compare |Z| with critical value

6.5 More worked examples

Problem 1 - single mean (two-tailed)

A random sample of 900 members has mean 3.4 cm. Can it be from a population with mean 3.25 cm and S.D. 2.61 cm? $Z = (3.4 - 3.25)/(2.61/30) = 0.15/0.087 = 1.72 < 1.96$ -> accept H0: sample is from that population.

Problem 2 - proportion (one-tailed)

A dealer claims at least 60 % of customers prefer brand X. In a sample of 100, 52 prefer X. H0: $P = 0.6$, H1: $P < 0.6$. $Z = (0.52 - 0.60)/\sqrt{0.6 \times 0.4/100} = -0.08/0.04899 = -1.633$. Critical (5 %) = -1.645. $|Z|$ smaller -> accept H0 (claim not rejected).

Problem 3 - difference of means

Two samples of sizes 400 and 600 have means 85 and 80 with S.D. 15 and 20 respectively. $Z = 5/\sqrt{225/400 + 400/600} = 5/\sqrt{0.5625 + 0.6667} = 5/1.1087 = 4.51 > 2.58$ -> significant difference.

Problem 4 - difference of proportions

In a town A, 50 of 400 persons are illiterate; in town B, 80 of 800. $p1 = 0.125$, $p2 = 0.10$; $\hat{P} = 130/1200 = 0.1083$. $S.E. = \sqrt{0.1083 \times 0.8917 \times (1/400 + 1/800)} = \sqrt{0.09658 \times 0.00375} = 0.01903$. $Z = 0.025/0.01903 = 1.31$ -> accept H0 at 5 %.

Problem 5 - confidence interval

Sample of 64 students has mean height 168 cm, S.D. 8 cm. 95 % C.I.: $168 \pm 1.96(1) = (166.04, 169.96)$. 99 % C.I.: $168 \pm 2.58 = (165.42, 170.58)$.

Problem 6 - sample size for proportion

To estimate a proportion within 3 % at 95 % confidence (no prior estimate, $p = 0.5$): $n = Z^2 p q / E^2 = (1.96)^2 (0.25)/(0.03)^2 = 0.9604/0.0009 = 1067$.

Problem 7 - decision table

Case	Z calc	Critical	Decision
1	2.1	1.96 (two-tailed 5 %)	Reject H0
2	-1.2	-1.645 (left 5 %)	Accept H0
3	2.4	2.33 (right 1 %)	Reject H0
4	1.8	2.58 (two-tailed 1 %)	Accept H0

Formula summary	Expression
S.E. of mean	sigma / sqrt(n)
S.E. of proportion	sqrt(PQ/n)
S.E. of (x1 - x2)	sqrt(s1^2/n1 + s2^2/n2)
Pooled proportion	(n1p1 + n2p2)/(n1 + n2)
Margin of error	Z(alpha/2) x S.E.
Sample size (mean)	n = (Z sigma/E)^2

Quick revision - Unit V

- Parameter (population) vs statistic (sample); S.E. of x-bar = sigma/sqrt(n)
- H0 (equality) vs H1; alpha = P(Type I); beta = P(Type II); power = 1 - beta
- Critical Z: 1.96 / 2.58 (two-tailed 5 % / 1 %); 1.645 / 2.33 (one-tailed)
- Z tests: single mean, two means, single proportion, two proportions (pooled P)
- C.I.: estimate +/- Z(alpha/2) x S.E.